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A Study of Effects of Digital Note Types on Long-Term Recall

Literature Review

Note-taking is a core skill for all university students. Being able to refer back to parts of a lecture that may have been forgotten is an indispensable advantage. In recent years, several different methods and frameworks for taking notes have come about; most importantly, there now exist digital applications such as Microsoft Word and Xournal that allow a student to digitally take and store notes (I. Salame et al., 2024). While each of these methods introduce advantages over longhand, they may also introduce drawbacks, such as distractions (A. Al-Sharman et al., 2025). Because digital note-taking is so prevalent to university courses and students, it is necessary to study the differences between digital note-taking styles.

Studies have consistently shown that note-taking, whether by hand or by typing, is an effective method of storing and memorizing information presented in a lecture. The precise framework for notes varies from person to person. J. Fisher and M. Harris (1974) found no overall significant difference in recall between students taking detailed notes, main points, or no notes. However, they did find that students assigned to take detailed notes had significantly better recall if they already preferred to take detailed notes. This result implies that students take most effectively when the exact method is left up to them. However, J. L. Austin et al. (2002) students who are given guided notes achieve higher recall on average, despite them not being able to choose for themselves. Therefore, some note structures may improve memory, even if the students do not prefer them. Finally, a survey from A. Al-

Sharman et al. (2025) found that students who habitually write in longhand have significantly better recall than those who use digital styluses. The exact effectiveness of note-taking depends on both the student's preference and the type of notes, with both having a strong effect on the ultimate effectiveness.

In addition to the general differences, other studies have also investigated the differences between handwriting and typing notes. A. Mangen et al. (2015) found that students who took notes by handwriting had higher recall than those who typed. They suggest that because handwriting requires much more precise and intricate hand motions, the words written may be better committed to memory. Results from P. Mueller and D. Oppenheimer (2014) add to this. They also found lower comprehension in students who use laptops; however, they suggest that because one can type much faster than handwriting, a student can record the lecture verbatim. So, they must interpret and paraphrase the information less, leading to lower comprehension. However, K. Morehead et al. (2019) repeated this study while also including students who take notes on eWriters and those who take no notes. They found small but non-significant differences between any of the groups, suggesting that any categorical comparison is premature. While many studies have found benefits to longhand note-taking, to say that longhand is always better would not necessarily be accurate.

Although these studies have compared different forms of handwritten notes to each other and to typed notes, little attention has been given to comparing different forms of typed notes to each other. Digital notes are not a monolith, and it is important to determine whether there is a difference between the types of digital notes. Many previous experiments grouped all forms of digital notes into one catch-all, not differentiating between the types. Some of the other studies were also not performed at an American private university like Carnegie Mellon. Thus the previous corpus of studies fails to investigate the differences between types of digital notes.

Research Proposal

Purpose

To address this gap in knowledge, we propose a study investigating the relationship, if one exists, between note-taking style and long-term knowledge retention. We will seek to answer this research question: “Does the structure by which a student takes typed notes affect their long-term memory and retention of the knowledge?” We hypothesize that more structured notes will be positively correlated to recall.

Participants

One hundred full-time first- and second-year undergraduate students will be selected across the colleges of Carnegie Mellon University; a convenience sample using public recruitment posters will be used to find and choose subjects. Students with excessive workloads, such as overloading, will be excluded, as they may not have the time or energy to adequately participate in this study. Students selected will be asked to commit to at most one hour of study for three non-consecutive weeks. As a reward, all participants will be gifted a \$15 gift card after the final session.

Procedures

Students will be randomly separated into three groups of thirty-three or thirty-four. The groups will be assigned to take notes by laptop in one of three methods: by bullet points, by Cornell notes, or by digital stylus. Participants will be provided the appropriate technology if need be, and the second group will be given a short explanation of Cornell notes beforehand. Each group will be shown a forty-five minute lecture on glacial flow, intended to be a topic that they are not familiar with, but that they should mostly understand on first watch. After the lecture, students’ notes will be collected, and they will be administered three short tests consisting of fifteen objective, multiple-choice questions from the lecture. They will not have access to their notes and will have a maximum of thirty minutes to complete it. The first of

these tests will take place immediately following the lecture. The second will take place one week later. The third will take place one month later. The three tests will only test the students' recall of different details from the lecture, and will be of similar difficulty.

Measurements

The tests will be separated into nine groups based on note-taking type and week. Each individual test will be scored based exclusively on the number of questions answered correctly. Their score will be the number of correct answers, ranging from zero to fifteen. The average and standard deviations for the test scores for each of the nine groups will be calculated. It is expected that the average scores across all types of note-taking will decline over time, due to the natural degradation of memory. Because we expect average scores to decline over time, averages across weeks will not be compared; only differences between groups of the same week will be directly tested. Two-sample t -tests will be used to determine whether there exists a statistically significant difference. If the more structured notes average score is significantly greater than other types of notes, then our hypothesis will be confirmed.

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